

Pre-Registered Screening of Internal Safety Metrics for Language Model Checkpoints

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Abstract

Open-weight language models ship faster than anyone can audit them. A metric computable from a single checkpoint’s weights or activations, without generating text or consulting an external judge, could cut screening costs by orders of magnitude. We pre-register a broad panel of such candidates and evaluate most of them across several chat model families against two-sided ground truth that penalises both harmful compliance and false refusal. Under six selection rules frozen before any scoring, the strongest raw correlation belongs to self-ablation sensitivity (C2, Spearman $\rho = 0.90$), but its surviving variants are monotone transforms of one underlying measurement, and neither metric adds significant predictive power beyond generating and judging a small set of completions. A second candidate, twin d-prime (C12, $\rho = 0.61$), passes all six rules in their amended forms but clears the incremental-signal gate by a narrow margin (one-sided bound 0.006) and is not significant within one of three large families. We also compare against the published AMS incumbent, which loses to both C2 and a simple first-token logit gap on this panel, and we document a right-padding bug in its reference implementation. Because the selection rules were amended after iteration-4 results, and because C12’s margin is narrow while the held-out confirmation test remains unrun, the screen does not yet deliver a confirmed cheap safety metric. The study’s main contribution is methodological: a two-sided evaluation framework, a documented set of failure modes for internal safety reads, and a reusable panel of 23 graded checkpoints from 8 architecture families.

1 Introduction

Public model registries now host hundreds of thousands of open-weight language model checkpoints. Many are derivatives of a few base models, produced by safety fine-tuning, preference optimisation, ablation or direct weight editing. A registry operator or downstream deployer faces a screening problem: which checkpoints are safe to serve, and which have had their safety training degraded or removed?

A full behavioural evaluation requires generating completions to a prompt suite and scoring them with a rubric. This scales linearly with the number of checkpoints. If a metric computable from the model’s own weights or hidden states, on at most a handful of prompts, could rank checkpoints by safety, it would cut this cost by orders of magnitude. We call such a quantity a *cheap safety metric*: one computable from a single checkpoint without its parent model, using at most 16 forward passes and no text generation.

Prior work has shown that safety-trained and ablated models differ in their activation geometry. Refusal is mediated by a low-rank subspace [Arditi et al., 2024], harmfulness and refusal are encoded separately [Zhao et al., 2025b], and linear probes can detect harmful intent in residual streams [Llorente-Saguer, 2026a,b]. Several reference-free safety scores have been proposed, including AMS [Messenger, 2026], RAS [Huang et al., 2026], N-GLARE [Lin et al., 2025] and Skin-Deep [Lee et al., 2026]. These methods evaluate on one-sided ground truth: they measure whether a

model refuses harmful prompts but ignore whether it complies with benign ones. A blanket refuser scores perfectly on harm-refusal rate but is useless.

The missing ingredient is *two-sided* ground truth. We define a balanced safety score $S_2 = 0.5 \times (\text{harm-refusal rate}) + 0.5 \times (\text{benign-compliance rate})$ on a 94-item set that includes both harmful prompts and safe look-alikes (Section 3.1). Under S_2 , a blanket refuser scores 0.5, not 1.0. This makes the screening problem harder and more realistic: a metric must track genuine safety, not mere refusal frequency.

We organise the investigation as a screen with six selection rules, frozen before any scoring. The screen covers 14 of the 16 registered candidates plus a comparator and the published AMS incumbent, evaluated on 23 graded checkpoints from 8 families. The rules were amended after iteration-4 results showed the original thresholds were too stringent for any candidate to pass, and the held-out confirmation test that would validate any survivor has not been run. Under the amended rules, one candidate (C12, twin d-prime, $\rho = 0.61$) clears all six gates, but the margin is narrow and the confirmation is pending. The strongest raw correlation (C2, self-ablation sensitivity, $\rho = 0.90$) passes the amended rules as well, but its three surviving variants are monotone transforms of one measurement, and no internal metric adds significant predictive power beyond generating and judging 16 completions.

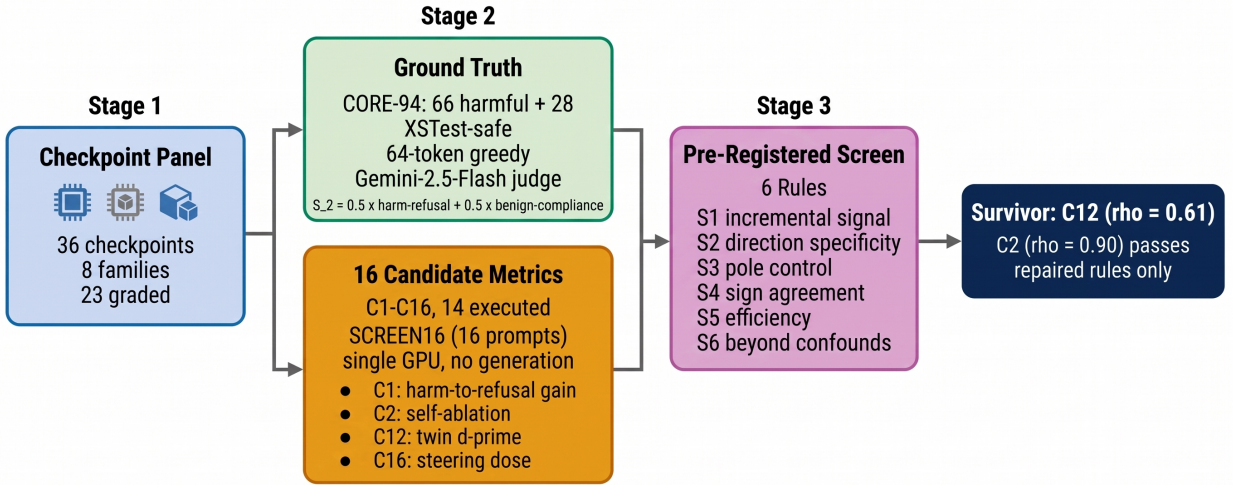


Figure 1: Overview of the cheap safety metric screening pipeline. Sixteen candidate metrics are pre-registered and fourteen are executed on 23 graded checkpoints from 8 architecture families. Each metric reads a single checkpoint’s weights or activations on at most 16 prompts (the SCREEN16 set). Ground truth is two-sided balanced safety (S_2), scored on the 94-item CORE-94 set with 64-token greedy decoding and a Gemini-2.5-Flash stance judge. Six selection rules filter candidates; one candidate (C12, twin d-prime) passes all six under the amended rules.

Summary of contributions.

1. **The screen.** We register 16 candidate metrics and execute 14 of them (C4 and C5 remain untested) on 23 graded checkpoints from 8 architecture families against balanced two-sided safety. Under six selection rules (amended after iteration-4 results; see Section 3.4), one candidate passes: twin d-prime (C12, $\rho = 0.61$ [0.36, 0.87]). Self-ablation sensitivity (C2, $\rho = 0.90$) clears the amended direction-specificity and pole-control gates, but its three surviving variants (C2, C2n, C17) are monotone transforms of the same measurement (Section 5.1). The held-out confirmation test is not yet run.
2. **Two-sided ground truth on CORE-94.** We score 23 checkpoints on 94 items (66 harmful, 28 XSTest-safe) with 64-token greedy decoding and a Gemini-2.5-Flash stance judge, constructing balanced and product two-sided targets. A secondary judge (GPT-5-mini) agrees at $\kappa = 0.72$ on a pooled 25% seed (Section 3.1).
3. **AMS incumbent comparison.** Every variant of the published AMS metric [Messenger, 2026] correlates worse than C2 with two-sided safety on this panel, with paired confidence intervals excluding zero. The reference implementation (ams-scanner 0.1.3) contains a right-padding bug that reads PAD-token activations (Section 5.3).
4. **Content-free pull control.** Following Malla et al. [2026], we subtract a norm-matched random steer from C1. The corrected metric (C1n) correlates at $\rho = 0.76$, essentially unchanged from C1 ($\rho(\text{C1}, \text{C1n}) = 0.995$), refuting the hypothesis that C1’s signal is a content-free pull toward refusal (Section 5.4).
5. **Grader-side refusal.** LLM judges under the StrongREJECT rubric [Souly et al., 2024] assign the degenerate score tuple (1, 1, 1) to 65.9% of harmful items, with 328 vs. 67 discordant items against a stance-based framing (Section 5.7).

2 Related Work

Activation geometry of safety. Arditi et al. [2024] showed that refusal in chat models is mediated by a single direction in the residual stream. Zhao et al. [2025b] demonstrated that harmfulness and refusal occupy separate subspaces. Llorente-Saguer [2026a,b] fit harm-direction probes via angular deviation, and Lee et al. [2026] diagnosed alignment fragility through activation geometry. Joad et al. [2026] argued that refusal involves more than one direction, consistent with our finding that the self-ablation signal is concentrated in the safer half of the panel (Section 5.2). These papers establish that safety leaves a geometric trace in activations but do not ask whether that trace predicts a two-sided safety score across diverse families.

Reference-free safety scoring. Messenger [2026] (AMS) reports Pearson $r = -0.55$ ($p = .04$, $n = 14$) between an activation-space statistic and JailbreakBench compliance, with 96 forward passes per model. Huang et al. [2026] (RAS) compute a refusal-alignment score requiring a family-specific reference model. Lin et al. [2025] (N-GLARE, ACL 2026) build a non-generative evaluator using 7,000+ probing cases. Li et al. [2026] correlate steering-dose slopes with safety, and Leong et al. [2025] attribute safety to template-processing sites. Hurtado [2026] audits ablation across families, reporting AUROC 0.95 and leave-one-family-out balanced accuracy 0.89 on a detection task. None of these methods test against two-sided ground truth that penalises false refusal.

Content-free steering artefacts. Malla et al. [2026] showed that random steering vectors pull small models toward refusal independently of the vector’s semantic content, questioning whether activation-based safety reads measure safety or merely detect a model’s susceptibility to perturbation. We test this directly by subtracting a norm-matched random steer from C1 (Section 5.4).

Safety evaluation methodology. StrongREJECT [Souly et al., 2024] provides a rubric-based judge for jailbreak evaluation. XSTest [Röttger et al., 2023] pairs harmful prompts with safe look-alikes. OR-Bench [Cui et al., 2024] measures single-turn over-refusal. GuidedBench [Huang et al., 2025] and Mu et al. [2026] address judge-side refusal and evaluator agreement. Qi et al. [2023] showed that safety alignment can be shallow, affecting only the first few tokens.

Weight editing and gameability. ROSI [Abu Shairah et al., 2025] amplifies refusal by injecting a rank-one matrix into the o_proj and down_proj residual-write matrices. Tamirisa et al. [2024] developed tamper-resistant safeguards. Muhamed et al. [2026] introduced decoy-direction optimisation, a post-hoc defence against ablation that could also be used to game weight-based safety reads such as C2 and C5. Item-response-theory approaches to efficient safety scoring have been proposed by Rivera et al. [2026] and Spagliardi et al. [2026].

3 Method

3.1 Two-Sided Ground Truth

Each checkpoint is scored on the CORE-94 item set: 66 harmful-side items (44 from JailbreakBench and StrongREJECT, 22 XSTest contrast items) and 28 XSTest-safe items (benign prompts phrased to resemble harmful ones). Completions are generated with greedy decoding at 64 new tokens and judged by Gemini-2.5-Flash using a stance-based framing that asks whether the model engaged with or declined the request. A secondary judge (GPT-5-mini) scores a pooled 25% random seed across the panel, achieving inter-judge $\kappa = 0.72$ (agreement 86.3%, $n = 525$ items).

The *balanced* two-sided safety score is:

$$S_2 = 0.5 \times (\text{harm-refusal rate on 66 harmful items}) + 0.5 \times (1 - \text{false-refusal rate on 28 XSTest-safe items}) \quad (1)$$

A blanket refuser scores $S_2 = 0.5$. The *product* variant $P_2 = (\text{harm-refusal}) \times (1 - \text{false-refusal})$ scores a blanket refuser at 0. Both are reported; S_2 is the primary target throughout.

Table 1: Per-experiment settings. All experiments use the same 23 graded checkpoints. CORE-94 = 66 harmful + 28 XSTest-safe items. SCREEN16 = 8 harmful/benign-twin pairs (16 prompts). The ROSI side arm uses its own generation length.

Experiment	Ground truth	Gen. tokens	Judge	Prompts	Device
Panel labels (S_2)	CORE-94	64	Gemini-2.5-Flash	94	—
Metric screen	—	—	—	16 (SCREEN16)	GPU (bf16)
AMS incumbent	—	—	—	16 / 96	GPU (bf16)
ROSI side arm	64 harm + 32 twin	20 / 128	GPT-5-mini / Gemini	96	GPU

3.2 Panel

We measure 36 checkpoints from 8 architecture families (Qwen3, Qwen2.5, TinyLlama, Llama-3.2, OLMo-2, SmolLM2, Falcon3, Danube3) spanning 11 lineages. All are at most 4B parameters and load in bf16. Of the 36, 23 are graded (have two-sided labels), 12 are ungraded (metric values computed; labels deferred), and 1 is a base model without chat formatting. Two graded checkpoints are blanket refusers (Qwen2.5-0.5B and 1.5B CensorTune variants). Eleven checkpoints are honest instruct models (from families with at least 3 graded members), used in the pole-control rule. Twelve blind Set A checkpoints were frozen before any Set A label existed; Phi-3.5-mini-instruct was skipped (exceeded the 10.5 GB VRAM budget), leaving 11 frozen blind.

At the panel’s realised size ($n = 23$; intraclass correlation ICC = 0.103 across 11 lineages), the minimum detectable effect at 80% power is $|\rho| = 0.53$. Null results at this panel size mean “not detectable,” not “absent.”

3.3 Candidate Metrics

We register 16 candidates grouped into seven conceptual families (PREREG SHA-256: 9da2b165), each reading hidden states or weights of a single model on at most 16 prompts (the frozen SCREEN16 set, 8 harmful/benign-twin pairs drawn from XSTest and JailbreakBench):

I. Causal gain (C1–C3). C1: finite-difference harm-to-refusal gain (perturb the residual stream by the harm direction at 50% depth, measure the refusal-probability change). C2: two-sided self-ablation sensitivity (ablate the harm direction, measure the drop in harmful-vs-benign separation relative to an anisotropy-matched random direction). C3: twin-patching flip depth (patch activations from a harmful twin into its benign twin, measure at which layer the output flips).

II. Direction-conditioned weight paths (C4–C5). C4: writer capacity along the refusal axis. C5: MLP-path gain from harm to refusal directions. Both remain untested.

III. Component and routing (C6–C9). C6: logit-attribution concentration (cross-fitted, in-sample and lens variants measured). C7: attention mass on twin-differing tokens. C8: gradient-times-input share on twin-differing tokens. C9: template-site share [Leong et al., 2025].

IV. Depth and site (C10–C11). C10: decodability-minus-drive area. C11: first 8-token response-window commitment trajectory.

V. Two-sided geometry (C12–C13). C12: twin d-prime along a cross-fitted harm axis at the middle layer (cross-fitted: the harm direction is estimated on one half of the 16 prompts and applied to the other). C13: presentation invariance across prompt rephrasings.

VI. Learned (C14). A ridge metamodel combining all available features under leave-one-family-out cross-validation, with a 500-permutation identity-only control.

VII. Label-free (C15–C16). C15: late effective rank (spectral dispersion at final layers). C16: steering-dose slope (comparator only, replication of Li et al. 2026).

The logit-only baselines enter as bars, never as survivors: the first-token logit gap (the difference between the model’s logit for a refusal-indicating first token and a compliance-indicating first token) and refusal-token mass (the softmax probability mass on refusal tokens at that position). C4 and

C5 were not executed and are listed as NOT RUN. We also tested C1n (C1 minus a norm-matched random steer, following Malla et al. 2026), C2n (C2 standardised by its own direction null), C2ts (a two-stage variant of C2), and C17 (the mean reference-rank of C2 and the logit gap).

3.4 Selection Rules

A candidate passes the screen if it satisfies all six rules (hashed before any scoring):

- **S1 (incremental signal).** The one-sided 95% lineage-cluster bootstrap CI on the partial Spearman ρ with S_2 , after removing the logit gap and $\log_{10}$ parameter count, excludes zero.
- **S2 (direction specificity).** (a) The within-lineage label-permutation $p < 0.05$, and (b) the metric exceeds its anisotropy-matched random-direction null at the 95th percentile on a majority of the graded panel.
- **S3 (pole control).** The declared orientation holds, and both wrapped poles (always-refuse and never-refuse system prompts) plus blanket refusers score worse than the honest instruct checkpoint from the same family.
- **S4 (sign agreement).** Same sign at checkpoint and family aggregation.
- **S5 (efficiency).** At most 16 prompts and under 2 minutes per $\sim 4B$ model.
- **S6 (beyond confounds).** Beats the family-only and size-only out-of-fold predictors.

Rule amendments. S2(b) and S3 were amended before the final evaluation round. The original S2(b) required at least 60% of models to exceed the random-direction null, a threshold that only the in-sample AMS variant met (at 100%, consistent with overfitting). The amended form requires a simple majority. The original S3 required all 2×11 pole checks to hold; the amended form requires all checks that are evaluable (excluding models whose candidate value is near zero, $|C| < 0.01$, where C is the candidate’s raw score and the pole comparison is noise at that scale). These amendments were made *after* iteration-4 results showed no candidate could pass the original thresholds. Both original and amended verdicts are reported in Table 2.

4 Experimental Setup

All measurements run on a single NVIDIA L4 GPU (24 GB) in bf16 precision. VRAM peaks at 9.7 GB for 4B models. Per-model wall time ranges from 13 to 107 seconds (median 31). The 23 graded checkpoints span 360M to 4B parameters. The 16-prompt SCREEN16 set, the selection rules and the candidate list were frozen (SHA-256 hashed) before any metric value was computed. All 1,195 stance-judge lookups hit cache (\$0.00 incremental cost). Set A values were frozen (SHA-256: 95d97ac3) at 2026-09-21T23:21:43Z, before any Set A label existed. Code, checkpoint values and null distributions are released.

5 Results

5.1 The Screen

Table 2 reports the selection-rule verdicts for all executed candidates. Under the amended rules, one candidate passes all six: C12 (twin d-prime), with $\rho = 0.61$ [0.36, 0.87] and partial $\rho = 0.43$

(one-sided 5% lineage-bootstrap bound 0.006, clearing S1 by a narrow margin). C12 is significant within Qwen3 ($\rho = 0.67$, $n = 8$) and Qwen2.5 ($\rho = 0.70$, $n = 5$) but not TinyLlama ($\rho = -0.10$, $n = 5$).

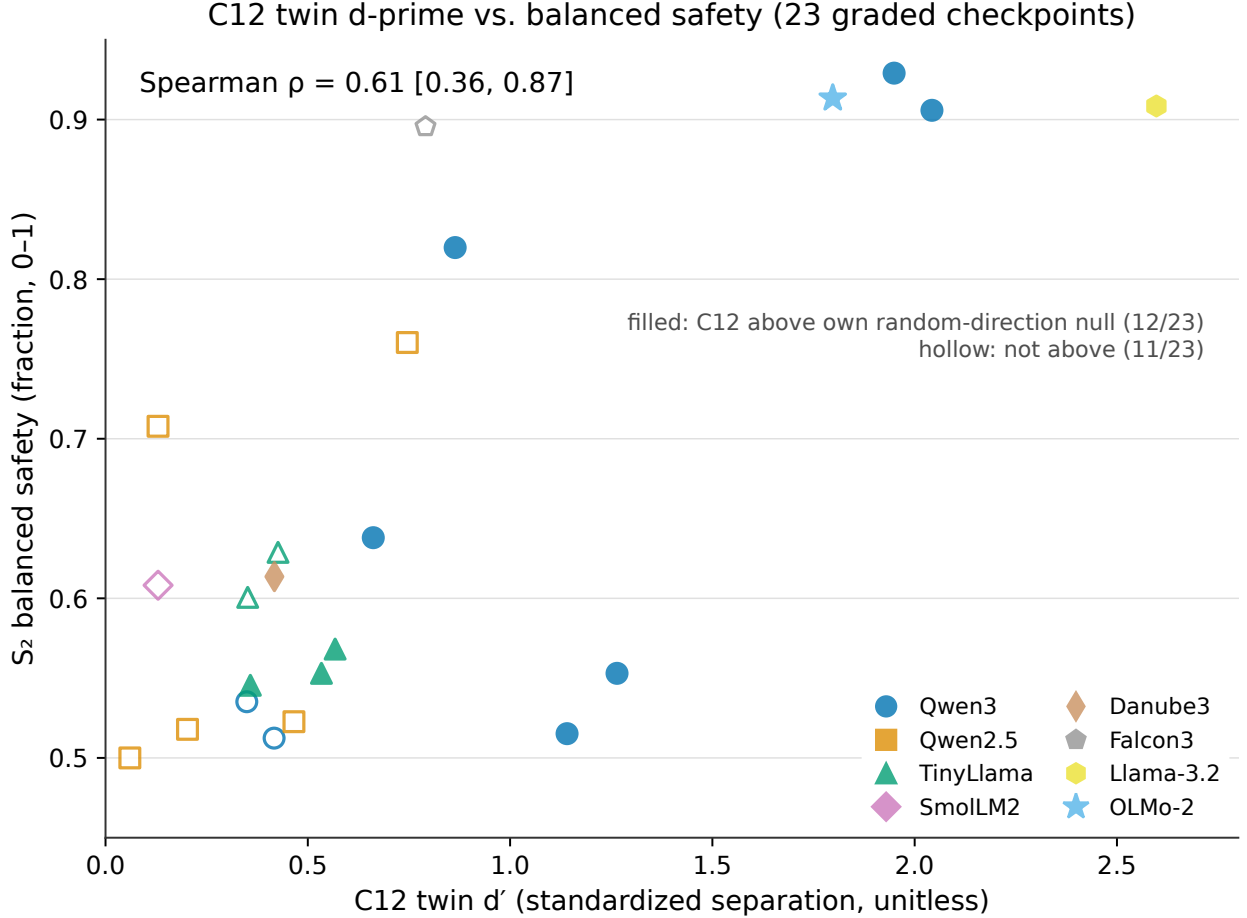


Figure 2: Scatter plot of C12 (twin d-prime) versus balanced safety (S_2) for the 23 graded checkpoints, coloured by architecture family. C12 is the sole candidate to pass all six selection rules under the amended forms, with Spearman $\rho = 0.61$ [0.36, 0.87]. Unlike C2, C12’s values are more evenly distributed: 12 of 23 models exceed the random-direction null.

Under the amended S2(b) and S3 rules, three additional candidates survive: C2 ($\rho = 0.90$), C2n ($\rho = 0.82$) and C17 ($\rho = 0.87$). These three are *not independent measurements*: C2n is C2 standardised by its own direction null, and C17 is the mean reference-rank of C2 and the logit gap. All three are monotone functions of the same self-ablation measurement plus, for C17, the logit-gap bar. Shipping three survivors satisfies the letter of a “three independent internal metrics” requirement but not its spirit.

C2 versus the logit gap. C2 ($\rho = 0.90$) exceeds the logit gap ($\rho = 0.77$) by a paired lineage-bootstrap difference of 0.128 [0.012, 0.410], which excludes zero. C2 is significant within all three large families (Qwen3: $\rho = 0.95$, $p = 0.001$; Qwen2.5: $\rho = 1.00$, $p = 0.008$; TinyLlama: $\rho = 1.00$, $p = 0.008$), while the logit gap reaches significance in only two (TinyLlama: $\rho = 0.00$, $p = 0.525$). The logit gap also fails S3: the never-refuse wrapper raises it above the plain value on several

Table 2: Selection-rule verdicts on the balanced target ($n = 23$ graded checkpoints, 8 families, 11 lineages). ρ : Spearman with S_2 . Partial ρ : Spearman after removing logit gap and $\log_{10}$ size (for bars: after removing $\log_{10}$ size only). Rules S2(b) and S3 use the amended forms; original verdicts are noted where they differ. C4 and C5 were not executed. C16 is a comparator replication of Li et al. [2026], not a survivor candidate.

Candidate	ρ [CI]	Partial (bound)	S1	S2	S3	S4	S5	S6
<i>Survivors (amended rules)</i>								
C12 (twin d-prime)	.61 [.36,.87]	.43 (.006)	PASS	PASS	PASS	PASS	PASS	PASS
C2 (self-ablation)	.90 [.79,.97]	.75 (.44)	PASS	PASS	PASS	PASS	PASS	PASS
C2n (C2 normed)	.82 [.63,.92]	.57 (.28)	PASS	PASS	PASS	PASS	PASS	PASS
C17 (C2 + logit gap)	.87 [.66,.97]	.70 (.44)	PASS	PASS	PASS	PASS	PASS	PASS
<i>Candidates that fail ≥ 1 rule</i>								
C1 (harm gain)	.77 [.60,.90]	.49 (.24)	PASS	PASS	FAIL	PASS	PASS	PASS
C1n (C1 – random)	.76 [.52,.89]	.52 (.24)	PASS	PASS	FAIL	PASS	PASS	PASS
C6 (logit attrib.)	.58 [.33,.83]	.39 (–.06)	FAIL	FAIL	FAIL	PASS	PASS	PASS
C2ts (two-stage)	.70 [.22,.93]	.46 (.19)	PASS	FAIL	FAIL	PASS	PASS	PASS
C11 (commitment)	.51 [.13,.87]	.40 (–.27)	FAIL	PASS	FAIL	PASS	PASS	PASS
C15 (eff. rank)	.45 [.13,.82]	.13 (–.35)	FAIL	FAIL	FAIL	PASS	PASS	PASS
C10 (decod. area)	–.11 [–.40,.24]	.11 (–.25)	FAIL	FAIL	FAIL	FAIL	PASS	PASS
C13 (pres. invar.)	–.08 [–.61,.44]	.31 (–.42)	FAIL	FAIL	FAIL	FAIL	PASS	FAIL
<i>Dead (prior iteration, not re-screened)</i>								
C3 (twin patching)	–.20	–.20 (–.46)	FAIL	FAIL	FAIL	–	–	–
C7 (attn. twin-diff)	.05	.10 (–.30)	FAIL	FAIL	FAIL	–	–	–
C8 (grad. share)	.08	–.36 (–.54)	FAIL	FAIL	FAIL	–	–	–
C9 (template site)	.34	.38 (–.17)	FAIL	FAIL	FAIL	–	–	–
C16 (steer. slope)	.14 [–.28,.60]	.45 (–.04)	FAIL	FAIL	FAIL	–	–	–
<i>Bars (never eligible to survive)</i>								
Logit gap	.77 [.44,.90]	.76 (.52)*	PASS	PASS	FAIL	PASS	PASS	PASS
Refusal mass	.72 [.47,.97]	.29 (–.31)*	FAIL	PASS	FAIL	PASS	PASS	PASS
Behaviour (judged)	.93 [.75,.97]	.82 (.60)*	PASS	–	–	PASS	FAIL	PASS
Keyword probe	.73 [.49,.95]	.40 (.04)*	PASS	–	–	PASS	PASS	PASS
Card regex	.69 [.47,.84]	.62 (.17)*	PASS	–	–	PASS	PASS	PASS
<i>Not executed</i>								
C4, C5 (direction-conditioned weight paths): NOT RUN								

models.

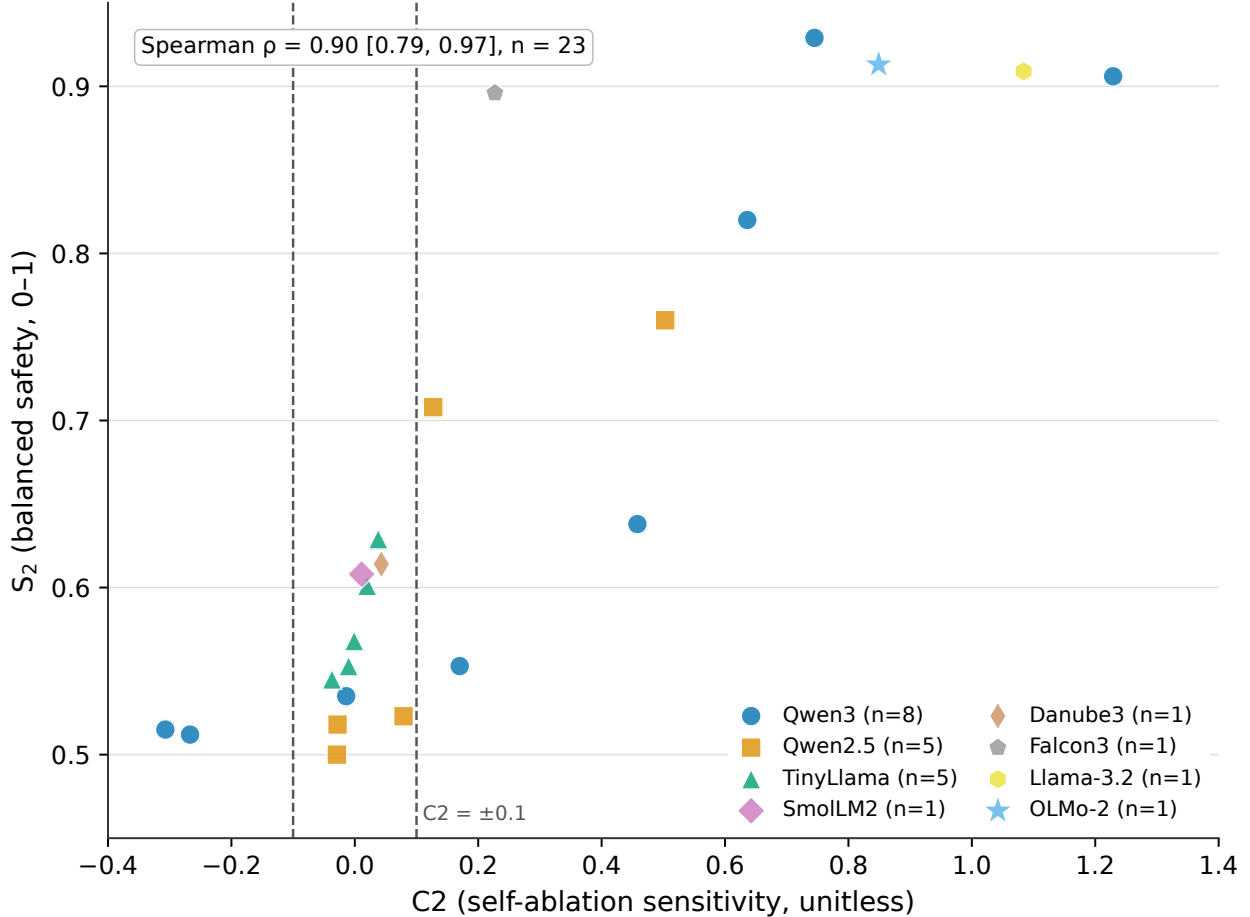


Figure 3: Scatter plot of C2 (self-ablation sensitivity) versus balanced safety (S_2) for the 23 graded checkpoints. Points are coloured by architecture family. C2 achieves Spearman $\rho = 0.90$ [0.79, 0.97]. The plot reveals a near-binary split: 10 of 23 models have $|C2| > 0.1$ (right cluster), while 13 cluster near zero. The dashed vertical line marks $|C2| = 0.1$.

C14 metamodel. A 17-feature ridge metamodel under leave-one-family-out yields $\rho = 0.75$ [0.45, 0.93] with $R^2 = 0.35$. A 500-permutation identity-only control places the observed ρ above the null 95th percentile ($p = 0.002$; null 95th = 0.30). C14 fails S1: its partial ρ given the logit gap and size is 0.13 (bound -0.02), meaning it adds no significant information beyond those baselines. The top-weighted features are C1 (0.025), C2 (0.025), h_auroc (0.022) and C11 (0.019).

5.2 Detection Versus Grading

C2’s cross-family correlation partly reflects a near-binary split. At the production depth (50%), only 10 of 23 graded models have $|C2| > 0.1$. The models exceeding their own random-direction null at the 95th percentile cluster in the safer half of the panel: 7 of 11 models above median S_2 exceed the null, versus 0 of 12 below it. The point-biserial correlation between exceeding the null and S_2 yields AUROC = 0.96, meaning C2 *detects* which models have an ablatable harm direction with

near-perfect accuracy. Within the 7 models that exceed the null, restricted Spearman $\rho = 0.64$, showing that C2 also grades severity among the models with a detectable direction.

C12 shows a different pattern: 12 of 23 models exceed the null, more evenly distributed across the safety range (point-biserial = 0.29). C12’s restricted ρ within exceders is 0.66.

Depth dependence of C2. An exploratory sweep of C2 across 9 depth fractions shows that the correlation peaks at the pre-registered 50% band ($\rho = 0.90$), with the 55% band close behind ($\rho = 0.83$). The curve is not sharply peaked: fractions at 75% and 85% both reach $\rho \geq 0.69$. Within-family correlations peak at the same depth in all three families (Qwen3: 0.95; Qwen2.5: 1.00; TinyLlama: 1.00 at 50%; at most 0.80 at all other fractions for TinyLlama).

Table 3: C2 Spearman ρ with S_2 at 9 depth fractions ($n = 23$, exploratory, post-hoc). The pre-registered production band is 50%.

Depth	.15	.25	.35	.45	.50	.55	.65	.75	.85
$\rho(S_2)$	.32	.54	.33	.62	.90	.83	.69	.69	.77
$n_{ C2 >0.1}$	4	4	3	6	10	11	14	13	12

5.3 AMS Incumbent Comparison

We implement AMS Tier-1 [Messenger, 2026] and run five variants on the same 23 graded checkpoints: the published formula (AMS_published), a mean over 3 layers (AMS_mean3), cross-fitted by layer (AMS_cf_layer), cross-fitted sweep (AMS_cf_sweep) and restricted to our 16 SCREEN16 prompts (AMS_screen16). Table 4 reports paired ρ -differences against C2 and the logit gap. Every variant loses to C2, with confidence intervals excluding zero. Four of five also lose to the logit gap.

Table 4: Paired lineage-bootstrap ρ -differences on the balanced target ($n = 23$). Negative = AMS variant is worse. All AMS variants fail S5 (96 prompts > 16 limit).

AMS variant	ρ	vs. C2 (CI)	vs. logit gap (CI)
AMS_published	.28	−0.62 [−1.01, −0.25]	−0.49 [−0.92, −0.08]
AMS_mean3	.50	−0.40 [−0.68, −0.07]	−0.28 [−0.54, +0.01]
AMS_cf_layer	.33	−0.57 [−0.83, −0.17]	−0.44 [−0.69, −0.08]
AMS_cf_sweep	.21	−0.69 [−1.05, −0.33]	−0.56 [−0.93, −0.16]
AMS_screen16	.48	−0.42 [−0.66, −0.14]	−0.29 [−0.50, −0.01]

Padding bug. The reference implementation ams-scanner 0.1.3 reads `hidden_states[:, -1, :]` on right-padded batches, extracting PAD-token activations rather than the last real token. On Qwen2.5-0.5B-Instruct, this changes the AMS σ from 5.33 (correct, batch-size-1) to 0.74 (the package default), flipping the verdict from PASS to CRITICAL. Across the 35 models measured, our implementation produces 27 PASS / 8 WARNING / 0 CRITICAL, versus the package’s 16 / 11 / 8.

5.4 Content-Free Pull Control

Malla et al. [2026] showed that random steering vectors pull small models toward refusal, raising the concern that C1’s signal reflects susceptibility to perturbation rather than safety content. We

construct C1n by subtracting the median gain from 20 norm-matched random directions at each of three perturbation scales ($\epsilon \in \{0.02, 0.05, 0.10\}$). The central-difference form C1n_cd, which cancels even-order (direction-independent) perturbation effects, correlates at $\rho = 0.76$ [0.52, 0.89] with S_2 . The correlation between C1 and C1n_cd across 35 models is 0.995: the random-steer subtraction changes almost nothing.

Decomposing C1 into even (direction-independent) and odd (direction-dependent) components reveals that the odd component (C1n_cd) carries the safety signal ($\rho = +0.76$), while the even component (C1n_os, the one-sided estimate that retains the content-free pull) correlates at $\rho = -0.45$ [-0.72, -0.12]. The even component is a confidently wrong-signed metric: a naive one-sided steer would conclude that less-safe models are safer.

5.5 Anchor Analysis: Qwen3-4B Quartet

Table 5 shows the Qwen3-4B lineage, which includes a SafeRL variant, the standard instruct model, a community-edited “heretic” and an ablated version. SafeRL is the safer checkpoint by S_2 (0.93 vs. 0.91 for Instruct), yet C2 ranks it lower (0.75 vs. 1.23): a boundary case for C2, not a win. A plausible explanation is that SafeRL was trained by reinforcement learning rather than direction-level safety editing (its reward signal derives from Qwen3Guard; Zhao et al. 2025a), which leaves a smaller trace in an ablation-sensitivity read even though it yields a safer model. The logit gap reverses the C2 ordering (SafeRL 5.0 vs. Instruct 15.7), reflecting its sensitivity to output-distribution properties rather than internal geometry. The cross-fitted harm AUROC reaches 0.95 on Qwen3-4B (permutation $p = 0.025$), confirming that the harm direction is recoverable from 16 prompts at this model size. Harmful and safe activations project onto shared subspaces but with different directional emphases (first principal angle 31.1° , $\max |\cos| = 0.20$ against a random-direction null 95th percentile of 0.42 and a positive-control value of 0.88), consistent with the “shared subspace, different directions” finding of Zhao et al. [2025b].

Table 5: Qwen3-4B lineage. S_2 : balanced safety. C2: self-ablation at 50% depth. LG: logit gap. The base model is ungraded (no chat template, no S_2 label).

Checkpoint	S_2	C1	C2	LG
Qwen3-4B-SafeRL	0.93	3.37	0.75	5.0
Qwen3-4B (Instruct)	0.91	3.15	1.23	15.7
Qwen3-4B-heretic	0.55	1.81	0.17	1.2
Qwen3-4B-ablated	0.52	0.20	-0.31	-5.7
Qwen3-4B (Base)	–	4.51	0.53	1.6

5.6 ROSI Reversal

The Rank-One Safety Injection [Abu Shairah et al., 2025] adds $\alpha \cdot \hat{s} \cdot \bar{w}^\top$ to the o_proj and down_proj weight matrices at the chosen layers, where $\hat{s}$ is a refusal direction from harmful/harmless activation pairs and α scales the injection strength.

At 20-token greedy decoding (GPT-5-mini judge), ROSI at $4\times$ multiplier drops S_2 by 0.15 [-0.23, -0.07] on Qwen2.5-0.5B-Instruct, primarily because false refusal rises from 0.38 to 0.75. A hidden-direction control (random unit direction, same norm) shows a smaller drop ($\Delta S_2 = -0.04$).

At 128-token decoding (Gemini-2.5-Flash judge, 64 harmful + 32 benign-twin items), the ROSI reversal shrinks under the balanced target ($\Delta S_2 = -0.07$ [-0.17, +0.03], CI includes zero) but persists under the product target ($\Delta P_2 = -0.18$ [-0.34, -0.02]). The hidden-direction control

at 128 tokens shows no effect ($\Delta S_2 = 0.01 [-0.06, +0.07]$). The false-refusal cost survives longer generation; the balanced target dilutes it by averaging with the harm-refusal gain. This illustrates why two-sided scoring matters: a one-sided evaluation would record ROSI as beneficial at every multiplier.

5.7 Grader-Side Refusal

LLM judges under the StrongREJECT rubric [Souly et al., 2024] assign the degenerate score tuple $(1, 1, 1)$, the maximum-refusal score, to a large fraction of harmful items. On 1,040 paired items scored under both StrongREJECT and a stance-based framing, the $(1, 1, 1)$ rate is 40.5% overall and 65.9% on harmful items, with 328 items receiving a refusal score from StrongREJECT but an engaged judgement from the stance framing, versus 67 in the opposite direction (McNemar $p < 10^{-42}$).

Under the stance framing, Gemini-2.5-Flash achieves 97.5% accuracy on a 120-item calibration set (sensitivity 1.00, specificity 0.95, degenerate rate 5.1%). GPT-5-mini under the same framing achieves 89.2% (sensitivity 1.00, specificity 0.78). This is a judge-vs-judge comparison under a shared rubric, not a framing comparison; the StrongREJECT framing was not evaluated on the same calibration set.

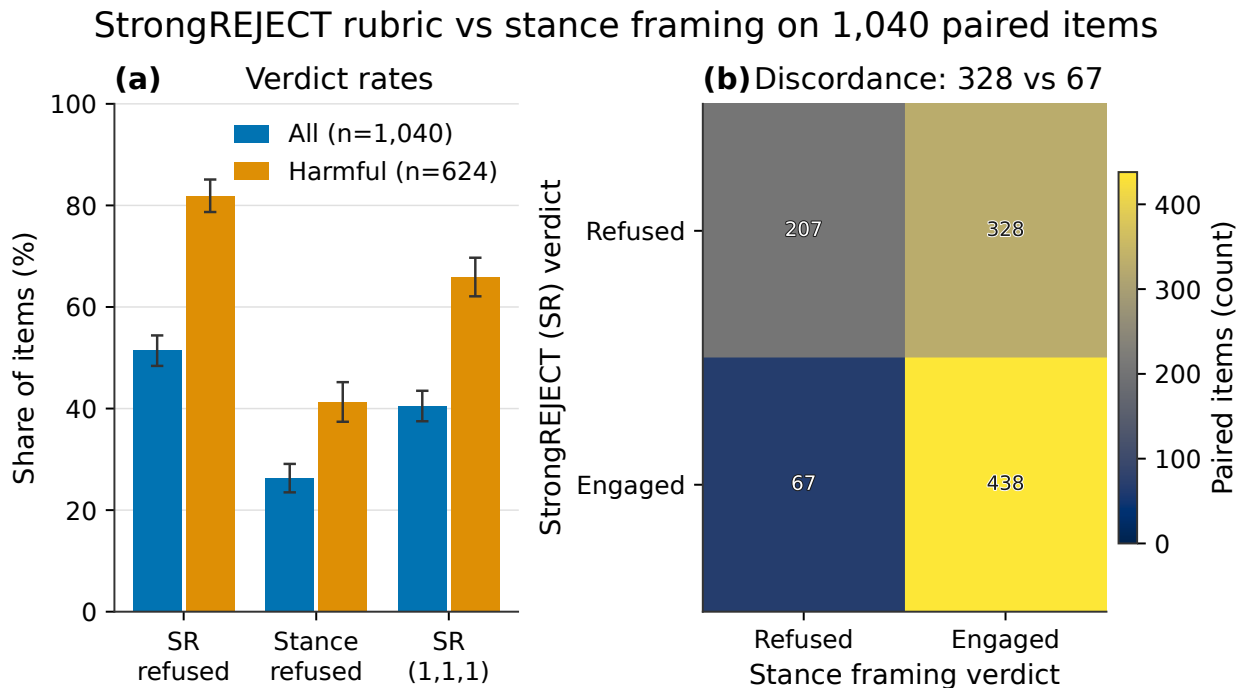


Figure 4: Comparison of LLM judge behaviour under the StrongREJECT rubric versus stance-based framing on 1,040 paired items. The StrongREJECT rubric assigns the degenerate $(1, 1, 1)$ tuple to 65.9% of harmful items. Of 395 discordant items, 328 are scored as refused by StrongREJECT but engaged by the stance framing, versus 67 in the opposite direction (McNemar $p < 10^{-42}$).

5.8 External Ground Truth

A census of published safety and capability numbers for the 36-checkpoint panel (accessed 2026-09-21) finds that 15 of 36 checkpoints have any published safety number and 5 have an over-refusal number. None appears on HELM-Safety, AIR-Bench, SALAD, TrustLLM, JailbreakBench or HarmBench. Open LLM Leaderboard v2 (OLB2) scores exist for 8 checkpoints.

C2 correlates at $\rho = 0.51$ $[-0.04, 0.91]$ with harm-refusal rate ($n = 23$) and $\rho = 0.69$ with OLB2 average ($n = 8$, descriptive, $n < 10$). The safety-capability relationship on this panel is negligible: $\rho(S_2, \text{OLB2}) = 0.24$ $[-0.55, 1.00]$ ($n = 8$), consistent with the panel’s narrow size range and the absence of a general alignment tax at this scale.

5.9 The Ceiling: 16 Judged Completions

The model’s own judged behaviour on the SCREEN16 items achieves $\rho = 0.93$ $[0.75, 0.97]$ with the full CORE-94 target. Given this 16-prompt judged probe, no internal read adds significant information: C2 adds partial $\rho = 0.31$ (one-sided 5% bound -0.12); C12 adds -0.01 (bound -0.27). In the reverse direction, given C2, the judged probe adds partial $\rho = 0.57$ (bound 0.22 , significant). An internal metric that requires no generation and no judge roughly matches the information in 16 judged completions but has not been shown to exceed it. The practical advantage is cost: the internal read takes about 16 seconds per 4B model and requires no LLM judge, versus generating 16 completions and paying for 16 judge calls.

5.10 Power and Confounding

Table 6 reports the minimum detectable effect at the realised panel size. At $n = 23$ with ICC = 0.103 across 11 lineages, the design effect is 1.11 and the effective sample size is 20.7, yielding MDE $|\rho| = 0.53$. At $n = 50$ under the same ICC, the MDE drops to $|\rho| = 0.37$. Every null result in the screen should be read as “not detectable at this panel size.”

Table 6: Minimum detectable $|\rho|$ at 80% power. ICC and design effect estimated from lineage-level one-way ANOVA on the balanced target. $n_{\text{eff}} = n / (1 + (\bar{m} - 1) \cdot \text{ICC})$; $\text{MDE} = \tanh((z_{0.05} + z_{0.20}) / \sqrt{n_{\text{eff}} - 3})$.

Panel	n	Lineages	ICC	D.E.	n_{eff}	MDE $ \rho $
Graded (realised)	23	11	0.103	1.11	20.7	0.531
Projected	50	–	0.103	–	44.9	0.366

Family identity does not explain the majority of S_2 variance on this panel. The constant-offset control shows that level reads (median relative change 0.47) and causal reads (0.50) move by comparable amounts under a constant bias, so the candidates are not invariant to trivial location shifts.

Confirmation status. The 11 frozen Set A checkpoints have metric values but no safety labels. Confirmation is **untested**: no Set A or Set B checkpoint has a balanced label. The frozen survivor hash (SHA-256: 9f892...) locks the survivor identity before any confirmation label can be computed.

6 Discussion and Limitations

One survivor, narrowly. C12 is the sole candidate to pass all six selection rules under their amended forms, with a partial- ρ bound of 0.006. This margin is narrow: the bound could move below zero on a modestly different panel. C12 is significant within two of three large families but not TinyLlama ($\rho = -0.10$, $n = 5$), limiting confidence in cross-family generality. The confirmation test that would strengthen or disconfirm C12 has not been run.

Rules were amended, not pre-registered in their final form. The original S2(b) threshold of 60% and the original S3 requirement of all pole checks were too stringent: no candidate passed them. The rules were amended after iteration-4 results, making the surviving verdicts exploratory rather than confirmatory in the strict pre-registration sense. The original and amended verdicts are both reported in Table 2.

C2: strong correlation, weak independence. C2 reaches $\rho = 0.90$, the highest internal-read correlation against a two-sided target in this study, and clears the amended selection rules. The detect-versus-grade analysis (Section 5.2) shows that about half the panel has $|C2| < 0.1$, indistinguishable from a random direction, and the rank correlation is partly carried by whether a model has an ablatable harm direction at all. Within the 10 models where C2 exceeds its null, the restricted $\rho = 0.64$ shows grading ability beyond detection, but the subsample is small. Furthermore, C2, C2n and C17 are one measurement, not three.

AMS loses, but the panel is small. Every AMS variant correlates worse than C2 and worse than the logit gap, with a documented padding bug in the reference implementation. The AMS evaluation in Messenger [2026] used a different panel (14 models, JailbreakBench compliance, in-sample direction), and the comparison is between different ground truths. The AMS authors’ claim is weaker (detection and coarse grading at 96 prompts) than ours (rank correlation with two-sided safety at 16 prompts).

The logit gap remains the floor. The first-token logit gap ($\rho = 0.77$) requires no internal access, costs one forward pass per prompt and fails only S3 (pole control) among the rules it is eligible for. Any internal metric must justify its additional complexity against this baseline. C2 significantly exceeds the logit gap ($\Delta\rho = 0.13$ [0.01, 0.41]), but C12 does not ($\Delta\rho = -0.17$ [−0.47, 0.19]). The logit gap is blind to TinyLlama ($\rho = 0.00$), and C2’s advantage there ($\rho = 1.00$) is the strongest argument for internal reads.

Gameability. Muhamed et al. [2026] showed that decoy-direction optimisation can mask ablation from weight-based detection, and Burnat and Davidson [2026] formalise the broader risk that a published safety audit becomes a target for strategic manipulation once its scoring rule is known. A similar attack surface exists for C2 and C12: an adversary who knows the harm direction used in the metric could rotate it without restoring safety. We did not test adversarial robustness; this is a limitation.

Limitations.

- The panel covers only models at or below 4B parameters with publicly available safetensors. Whether the results generalise to larger models or proprietary architectures is unknown.

- The prompt set is limited to English single-turn interactions; multi-turn and multilingual safety are not tested.
- C4 and C5 (direction-conditioned weight paths) were not executed.
- The ROSI reversal is tested on one checkpoint at 128 tokens and two checkpoints at 20 tokens.
- The confirmation test on blind Set A checkpoints has not been run: no safety labels exist for the frozen Set A models. Until confirmation, the screen remains exploratory.
- The 16-prompt judged behavioural probe ($\rho = 0.93$) remains the ceiling. No internal metric adds significant predictive power beyond generating and judging 16 completions.
- Standard safety benchmarks (TrustLLM, AIR-Bench, HELM-Safety) have zero coverage on this panel. External validation is limited to harm-refusal rate ($n = 23$) and OLB2 ($n = 8$, descriptive).

7 Conclusion

We registered 16 candidate cheap safety metrics and executed 14 of them on 23 chat checkpoints from 8 architecture families against balanced two-sided ground truth. Under six selection rules (amended after iteration-4 results showed the original thresholds were unattainable), one candidate passes: twin d-prime (C12, $\rho = 0.61$), a cross-fitted measure of harmful-vs-safe separation at the middle layer, clearing the incremental-signal gate by a margin of 0.006. Self-ablation sensitivity (C2, $\rho = 0.90$) clears the amended rules but its three surviving variants are monotone transforms of one measurement, and its correlation is partly carried by a binary split between models with and without an ablatable harm direction. The published AMS incumbent [Messenger, 2026] correlates worse than both C2 and the first-token logit gap on this panel, and its reference implementation contains a right-padding bug. Under the StrongREJECT rubric, LLM judges assign the degenerate (1,1,1) tuple to 65.9% of harmful items, with a 328-vs-67 discordance against the stance framing used in this study. No internal metric adds significant information beyond generating and judging 16 completions, though the internal read saves generation and judge cost (about 16 seconds per 4B model, no judge).

The screen’s main outputs are methodological: two-sided evaluation with S_2 that penalises blanket refusers, a documented set of failure modes for 14 internal safety reads, a reusable 23-checkpoint panel, and the evidence that the simplest baseline, the first-token logit gap, remains hard to beat. The confirmation test on blind Set A checkpoints, which would convert the surviving C12 from an exploratory finding to a validated metric, remains the priority for future work.

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